A Phenomenological Structural Selection Measure for String Backgrounds

Comparison with Swampland Constraints, No-Boundary Weights, and Anthropic Conditioning

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Abstract

String theory provides a candidate dynamical framework for quantum gravity, but its landscape of admissible backgrounds leaves open a separate question of realisation: among many consistent compactifications or cosmological histories, which backgrounds are structurally preferred? Structural Selection Theory proposes that this question should be treated by an additional selection functional, not by dynamics alone.

This paper develops a cautious phenomenological version of that idea. It does not claim a completed Theory of Everything, a unique Standard Model compactification, or a solution of the cosmological-constant problem. Instead, it addresses one concrete missing step: the relation between structural selection and existing vacuum-selection ideas, including Swampland constraints, Hartle–Hawking or no-boundary weights, and anthropic conditioning.

The central result is a combined phenomenological measure on a regulated background ensemble:

$$\boxed{\mathrm{d}\mathbb{P}_{\mathrm{pheno}}(\mathcal{B} \mid \mathcal{D}) = \frac{1}{Z_{\mathcal{D}}}\, \chi_{\mathrm{adm}}[\mathcal{B}]\, \exp[-\mathcal{F}_{\mathrm{sel}}[\mathcal{B}] - \mathcal{F}_{\mathrm{SW}}[\mathcal{B}]]\, W_{\cos}[\mathcal{B}]\, \mathcal{L}_{\mathrm{obs}}(\mathcal{D} \mid \mathcal{B})\, \mathrm{d}\mu_{\mathrm{scan}}[\mathcal{B}].} \quad (1)$$

Here μ_{scan} is the scan or construction measure, χ_{adm} is a hard admissibility filter, $\mathcal{F}_{\mathrm{SW}}$ encodes soft Swampland-like penalties, $W_{\cos}$ represents cosmological or no-boundary weighting, $\mathcal{L}_{\mathrm{obs}}$ encodes phenomenological and anthropic conditioning, and $\mathcal{F}_{\mathrm{sel}}$ is the structural selection functional

$$\mathcal{F}_{\mathrm{sel}}[\mathcal{B}] = -\sum_{a\in\{H,B,S,V,R,\Sigma\}} \lambda_a \log(\epsilon + \Gamma_a[\mathcal{B}]), \qquad \Gamma_a[\mathcal{B}] = \frac{1}{1 + d_a[\mathcal{B}]}.$$

The resulting architecture cleanly separates exclusion, prior weighting, structural ranking, and observational conditioning. It turns the landscape problem into a testable comparison between rankings rather than a purely philosophical claim.

The numerical part of the paper progressively instantiates this measure through five benchmarks: a 10D-near tadpole toy model, a flux-to-KKLT bridge, an LCS period-to-KKLT bridge, a Picard–Fuchs mirror-quintic period bridge, and a final backreaction/Standard-Model-sector proxy test. These layers are then consolidated into a unified phenomenological master functional.

1 Purpose and Status

This paper is deliberately phenomenological. Its purpose is not to derive the full ten- or eleven-dimensional theory, not to construct the observed Standard Model vacuum, and not to solve the cosmological constant from first principles. Its purpose is narrower:

> *to define how structural selection can be combined with existing string-landscape selection ideas in one explicit measure.*

This is an important intermediate step. A full microscopic derivation would require the exact reference measure on the space of admissible string backgrounds, a complete compactification ensemble, and a controlled connection to four-dimensional phenomenology. Those ingredients are not yet available in general. Nevertheless, one can already ask a mathematically precise question:

If a regulated ensemble is given, how should different selection principles be combined and compared?

The answer proposed here is a layered measure. Each existing idea is assigned a specific role:

- Swampland criteria act primarily as exclusion rules or consistency penalties.
- Hartle–Hawking or no-boundary proposals act as cosmological weighting of histories.
- Anthropic arguments act as conditional likelihoods.
- Structural selection acts as a pre-anthropic structural prior.

This separation prevents category mistakes. A veto is not a probability measure; an observational likelihood is not a dynamical law; a cosmological wavefunction is not automatically a structural stability criterion. The proposal is that these ingredients can be combined, but they should not be confused.

2 Background Ensemble

Let $\mathcal{B}$ denote a candidate string background. Depending on the ensemble, it may include a ten-dimensional or eleven-dimensional geometry, compact manifold, fluxes, branes, orientifolds, gauge bundles, moduli, low-energy effective fields, and cosmological initial data.

Let

$$\mathfrak{B}_{\text{scan}}$$

be a regulated ensemble generated by a specific construction or numerical scan. The measure μ_{scan} is not assumed to be fundamental. It is the reference measure associated with the chosen construction. Examples include:

- a flux scan in type IIB compactifications,
- an ensemble of effective KKLT-like vacua,
- a set of Calabi–Yau compactifications with bundle data,
- a cosmological history ensemble weighted by semiclassical amplitudes.

The present paper does not attempt to solve the measure problem. Instead, it makes the dependence on the reference ensemble explicit. All claims are therefore conditional:

$$\text{ranking} = \text{ranking relative to } (\mathfrak{B}_{\text{scan}}, \mu_{\text{scan}}).$$

3 Structural Selection as a Prior

The structural selection layer is encoded by a defect functional. For each background $\mathcal{B}$, define six non-negative dimensionless defects:

$$d_a[\mathcal{B}] \geq 0, \qquad a \in \{H, B, S, V, R, \Sigma\}.$$

They represent:

- H: equation or beta-function residuals,
- B: Bianchi, tadpole, anomaly, and constraint closure defects,
- S: controlled dynamical activity or excitation defects,
- V: inter-sector coupling and correlation defects,
- R: stability and robustness defects,
- Σ: global geometric and effective-field-theory integrity defects.

Each defect is converted into a bounded support factor:

$$\Gamma_a[\mathcal{B}] = \frac{1}{1 + d_a[\mathcal{B}]}, \qquad 0 < \Gamma_a \leq 1.$$

The structural selection functional is then

$$\boxed{\mathcal{F}_{\text{sel}}[\mathcal{B}] = -\sum_{a \in \{H,B,S,V,R,\Sigma\}} \lambda_a \log(\epsilon + \Gamma_a[\mathcal{B}]),} \tag{2}$$

with $\lambda_a \geq 0$ and $0 < \epsilon \ll 1$.

The associated structural prior is

$$\mathrm{d}\mathbb{P}_{\text{SST}}[\mathcal{B}] = \frac{1}{Z_{\text{SST}}} \exp[-\mathcal{F}_{\text{sel}}[\mathcal{B}]] \mathrm{d}\mu_{\text{scan}}[\mathcal{B}]. \tag{3}$$

This prior is not anthropic. It does not ask whether observers arise. It asks whether a background is structurally coherent according to equation closure, constraint closure, controlled activity, coupling organisation, robustness, and global integrity.

4 A 10D Covariant MAAT Support Functional

The symbolic MAAT expression

$$\text{ToE}_{\text{MAAT}} = \int \frac{(H + B + S + V + R)\, Z\, T_{\text{s}}}{\Delta E + \Delta Q + M_{\text{dim}}} \, \mathrm{d}t$$

is useful as an intuition, but it cannot be used literally as a ten-dimensional string-theoretic object. The time integral must be replaced by a covariant integral over a regulated ten-dimensional background, the string tension must carry the correct dimensional weight, and all numerator and denominator terms must be made scalar and dimensionless.

The corresponding 10D support functional is therefore defined as

$$\boxed{\mathcal{M}_{10}^{\text{MAAT}}[\mathcal{B}] = \frac{1}{\mathcal{V}_{10}} \int_{\mathcal{M}_{10}} \mathrm{d}^{10}x \sqrt{|G_{10}|}\, T_{\text{s}}^5\, Z_{\text{st}}(x;\mathcal{B}) \frac{\mathcal{N}_{\text{MAAT}}(x;\mathcal{B})}{\varepsilon + \Delta_E^{(10)}(x;\mathcal{B}) + \Delta_Q^{(10)}(x;\mathcal{B}) + \Delta_{\text{dim}}^{(10)}(x;\mathcal{B})}.} \tag{4}$$

Here $T_{\text{s}} = (2\pi\alpha')^{-1}$ and the factor $T_{\text{s}}^5 \sim \ell_s^{-10}$ makes the ten-dimensional volume integral dimensionless. The normalisation

$$\mathcal{V}_{10} = \int_{\mathcal{M}_{10}} \mathrm{d}^{10}x \sqrt{|G_{10}|}\, T_{\text{s}}^5$$

is understood on the same regulated domain or Euclidean saddle. The absolute value in $\sqrt{|G_{10}|}$ indicates that the expression is a support measure, not a Lorentzian action phase.

The local MAAT numerator is the bounded structural support

$$\mathcal{N}_{\mathrm{MAAT}}(x;\mathcal{B}) = \frac{H_{10}(x) + B_{10}(x) + S_{10}(x) + V_{10}(x) + R_{10}(x)}{5}, \qquad 0 \le \mathcal{N}_{\mathrm{MAAT}} \le 1. \tag{5}$$

Each sector is constructed from a defect through

$$X_{10}(x) = \frac{1}{1 + d_X^{(10)}(x)}, \qquad X \in \{H, B, R\},$$

or, for activity and connectivity, through bounded support functions that reward controlled excitation and coherent coupling rather than unbounded growth.

The five symbolic MAAT fields acquire the following 10D interpretation:

- H_{10} measures equation or worldsheet beta-function closure, for example through residuals of β^G_{MN}, β^B_{MN}, β^Φ, or the ten-dimensional supergravity equations.
- B_{10} measures Bianchi identities, tadpole closure, anomaly cancellation, and brane/source consistency.
- S_{10} measures controlled dynamical activity, such as nontrivial but non-runaway flux, curvature, or moduli excitation.
- V_{10} measures coherent interconnection between sectors, such as flux–geometry, brane–moduli, or gauge–geometry coupling structure.
- R_{10} measures robustness, including absence of tachyonic instabilities outside allowed windows, metastability margins, and distance from singular limits.

A concrete defect closure can be written schematically as

$$d_H^{(10)} = \frac{\|\beta^G\|^2}{\Lambda_G^2} + \frac{\|\beta^B\|^2}{\Lambda_B^2} + \frac{\|\beta^\Phi\|^2}{\Lambda_\Phi^2} + \cdots, \tag{6}$$

$$d_B^{(10)} = \sum_p \frac{\|\mathrm{d}F_p - H_3 \wedge F_{p-2} - j_p\|^2}{\Lambda_{B,p}^2} + \sum_i \left(\frac{\Delta Q_i}{Q_i^{\mathrm{scale}} + 1}\right)^2, \tag{7}$$

$$d_R^{(10)} = \sum_i [-m_i^2 L^2]_+^2 + d_{\mathrm{sing}} + d_{\mathrm{decay}}. \tag{8}$$

The constants $\Lambda_G, \Lambda_B, \Lambda_\Phi, \Lambda_{B,p}$ are normalisation scales fixed by the chosen ensemble. This keeps the construction dimensionless and makes clear that the present paper defines an effective closure, not a unique microscopic derivation.

The denominator in Eq. (4) separates three obstructions:

$$\Delta_E^{(10)} = \text{dimensionless excess energy/backreaction cost}, \tag{9}$$

$$\Delta_Q^{(10)} = \sum_i \left|\frac{\Delta Q_i}{Q_i^{\mathrm{scale}} + 1}\right|, \tag{10}$$

$$\Delta_{\mathrm{dim}}^{(10)} = (d_{\mathrm{mac}} - 4)^2 + \left(\frac{\ell_s}{L_{\mathrm{comp}}}\right)^2 + \left(\frac{H_4}{m_{\mathrm{KK}}}\right)^2 + d_{\mathrm{offdiag}}. \tag{11}$$

Thus M_{dim} in the symbolic formula becomes a concrete dimensional-reduction defect. It penalises backgrounds that fail to produce a controlled four-dimensional effective world, lack scale separation, or contain large off-diagonal mixing between external and compact directions.

Finally, the string partition factor Z must also be normalised. The raw worldsheet partition function may be divergent or prescription-dependent, so the ten-dimensional expression uses a bounded support factor

$$0 \le Z_{\mathrm{st}}(x;\mathcal{B}) \le 1,$$

which may encode modular consistency, one-loop control, or a renormalised background weight. In the most conservative implementation one sets $Z_{\rm st} = 1$ and lets the remaining diagnostics carry the ranking.

The structural selection functional associated with the 10D MAAT support is

$$\boxed{\mathcal{F}^{(10)}_{\rm sel}[\mathcal{B}] = -\log\big(\varepsilon + \mathcal{M}^{\rm MAAT}_{10}[\mathcal{B}]\big)\,.} \tag{12}$$

Equation (12) does not solve the full ten-dimensional problem, but it gives it a precise target form: a candidate 10D background is not merely checked for consistency; it is assigned a covariant, dimensionless structural support score. A complete derivation would require computing the defects directly from a specified string compactification ensemble.

5 Swampland Criteria: Exclusion versus Ranking

Swampland criteria are best interpreted as quantum-gravity consistency constraints. Their natural mathematical form is a filter:

$$\chi_{\rm SW}[\mathcal{B}] = \prod_i \mathbf{1}[c_i(\mathcal{B}) \le 0]\,,$$

where each $c_i(\mathcal{B}) \le 0$ represents a proposed consistency condition.

This hard form is appropriate when the criterion is treated as exact. However, many Swampland conjectures are still under debate, may admit refined versions, or may be difficult to evaluate exactly in a reduced effective model. For phenomenological work it is useful to introduce a soft penalty:

$$\mathcal{F}_{\rm SW}[\mathcal{B}] = \sum_i \eta_i\,[c_i(\mathcal{B})]_+^2, \qquad [x]_+ = \max(x,0). \tag{13}$$

The hard-veto limit is recovered as $\eta_i \to \infty$. Thus Swampland input can be included either as:

$$\chi_{\rm adm}[\mathcal{B}] \quad \text{or} \quad \exp[-\mathcal{F}_{\rm SW}[\mathcal{B}]].$$

The conceptual distinction from structural selection is important:

Swampland: is the EFT allowed by quantum gravity?

Structural selection: among allowed candidates, which are favoured?

The two ideas are therefore complementary, not redundant.

6 No-Boundary and Cosmological Weights

Hartle–Hawking and related no-boundary proposals assign semiclassical weights to cosmological histories through Euclidean or complex saddle contributions. Schematically, one may write

$$W_{\rm cos}[\mathcal{B}] \sim \exp[-2\,{\rm Re}\, I_E(\mathcal{B})]$$

or another prescription-dependent amplitude weight. The precise sign, contour, and saddle structure are subtle and proposal-dependent; the present paper does not attempt to settle those issues.

For the phenomenological architecture, the key point is role separation:

$$W_{\rm cos}[\mathcal{B}] \quad \text{weights cosmological histories,}$$

whereas

$$\mathcal{F}_{\rm sel}[\mathcal{B}] \quad \text{ranks structural coherence.}$$

These two contributions can be combined in a measure without identifying them:

$$\mathrm{d}\mathbb{P}[\mathcal{B}] \propto \exp[-\mathcal{F}_{\mathrm{sel}}[\mathcal{B}]]\, W_{\cos}[\mathcal{B}]\, \mathrm{d}\mu_{\mathrm{scan}}[\mathcal{B}].$$

If a future derivation showed that structural defects arise directly inside the semiclassical action, then $\mathcal{F}_{\mathrm{sel}}$ could become part of the fundamental cosmological weight. Until then, the cautious interpretation is that structural selection supplies an additional effective prior over backgrounds.

7 Anthropic Conditioning

Anthropic reasoning is not a prior over fundamental structures. It is a conditioning step:

$$\mathbb{P}(\mathcal{B} \mid \mathcal{D}) \propto \mathbb{P}(\mathcal{D} \mid \mathcal{B})\, \mathbb{P}(\mathcal{B}).$$

In the present notation, observational and anthropic conditioning is written as a likelihood:

$$\mathcal{L}_{\mathrm{obs}}(\mathcal{D} \mid \mathcal{B}) = \mathcal{L}_{\mathrm{LE}}(D_{\mathrm{LE}} \mid \mathcal{B})\, \mathcal{L}_{\Lambda}(D_{\Lambda} \mid \mathcal{B})\, \mathcal{L}_{\cos}(D_{\cos} \mid \mathcal{B})\, \mathcal{L}_{\mathrm{anth}}(D_{\mathrm{anth}} \mid \mathcal{B}).$$

This makes the separation sharp:

- $\mathcal{F}_{\mathrm{sel}}$ is a structural prior,
- $\mathcal{F}_{\mathrm{SW}}$ is a consistency penalty,
- $W_{\cos}$ is a cosmological weighting,
- $\mathcal{L}_{\mathrm{obs}}$ is a phenomenological or anthropic likelihood.

Thus anthropic selection does not disappear, but it is no longer asked to carry the whole burden of vacuum selection.

8 Phenomenological Master Measure

Combining the preceding ingredients gives the central formula of this paper:

$$\boxed{\mathrm{d}\mathbb{P}_{\mathrm{pheno}}(\mathcal{B} \mid \mathcal{D}) = \frac{1}{Z_{\mathcal{D}}}\, \chi_{\mathrm{adm}}[\mathcal{B}]\, \exp[-\mathcal{F}_{\mathrm{sel}}[\mathcal{B}] - \mathcal{F}_{\mathrm{SW}}[\mathcal{B}]]\, W_{\cos}[\mathcal{B}]\, \mathcal{L}_{\mathrm{obs}}(\mathcal{D} \mid \mathcal{B})\, \mathrm{d}\mu_{\mathrm{scan}}[\mathcal{B}].} \tag{14}$$

Equivalently, for ranking purposes one may define the negative log-score

$$\boxed{\mathcal{F}_{\mathrm{pheno}}[\mathcal{B} \mid \mathcal{D}] = \mathcal{F}_{\mathrm{sel}}[\mathcal{B}] + \mathcal{F}_{\mathrm{SW}}[\mathcal{B}] - \log W_{\cos}[\mathcal{B}] - \log \mathcal{L}_{\mathrm{obs}}(\mathcal{D} \mid \mathcal{B}) - \log \chi_{\mathrm{adm}}[\mathcal{B}],} \tag{15}$$

where $\chi_{\mathrm{adm}} = 0$ is interpreted as an infinite penalty.

This is the cleanest phenomenological statement of the proposal:

$$\text{selected candidates} = \arg \min_{\mathcal{B} \in \mathfrak{B}_{\mathrm{scan}}} \mathcal{F}_{\mathrm{pheno}}[\mathcal{B} \mid \mathcal{D}].$$

9 Cosmological Constant as Conditioning, Not Yet as Solution

The observed cosmological constant is one of the hardest selection problems. This paper does not solve it. It only defines how it enters the combined measure.

A simple Gaussian conditioning likelihood is

$$\mathcal{L}_\Lambda(D_\Lambda \mid \mathcal{B}) = \exp\left[-\frac{(\Lambda_{\rm eff}[\mathcal{B}] - \Lambda_{\rm obs})^2}{2\sigma_\Lambda^2}\right]. \tag{16}$$

Alternatively, an environmental window may be used:

$$\mathcal{L}_\Lambda(D_\Lambda \mid \mathcal{B}) = \mathbf{1}[\Lambda_{\rm min} \le \Lambda_{\rm eff}[\mathcal{B}] \le \Lambda_{\rm max}].$$

The important point is that Λ-conditioning is not identical to structural selection. A background can have a favourable cosmological constant while being structurally fragile, and a structurally coherent background can still fail observational conditioning. The combined measure keeps these facts separate.

10 Minimal Ranking Protocol

The proposal becomes testable once a finite ensemble is specified. For a set

$$\mathfrak{E} = \{\mathcal{B}_1, \ldots, \mathcal{B}_N\},$$

the following protocol compares selection schemes:

1. Compute ordinary vacuum data, such as $\Lambda_{\rm eff}$, stationarity residuals, mass spectra, and EFT-control quantities.
2. Evaluate hard admissibility filters $\chi_{\rm adm}$.
3. Evaluate soft Swampland penalties $\mathcal{F}_{\rm SW}$ where applicable.
4. Compute structural defects d_a and the structural score $\mathcal{F}_{\rm sel}$.
5. Compute observational likelihoods $\mathcal{L}_{\rm obs}$.
6. Compare four rankings:

$$\mathrm{Rank}_E, \qquad \mathrm{Rank}_{\rm SW}, \qquad \mathrm{Rank}_{\rm anth}, \qquad \mathrm{Rank}_{\rm pheno}.$$

The proposal is nontrivial only if $\mathrm{Rank}_{\rm pheno}$ is neither identical to energy ranking nor reducible to any single existing criterion. This gives an immediate falsification route:

$\mathcal{F}_{\rm sel}$ is useful only if it adds stable ranking information.

11 First Minimal Test: A 10D-Near Tadpole Toy Ensemble

We now implement the first deliberately small test of the 10D construction. The goal is not to scan realistic Calabi–Yau compactifications. The goal is to make one sector operational: the B_{10} balance sector associated with Bianchi identities, charge closure, and tadpole cancellation.

We generate a type-IIB-inspired flux ensemble with integer flux vectors $F, H \in \mathbb{Z}^8$ and a symplectic D3-tadpole proxy

$$N_{\rm flux} = F^T J H. \tag{17}$$

For a fixed tadpole bound $L = 24$, each candidate is assigned an integer brane number N_{D3} and a charge residual

$$\Delta Q_{D3} = N_{\text{flux}} + N_{D3} - L. \tag{18}$$

The operational B_{10} defect is

$$d_B^{(10)} = 18 \left(\frac{\Delta Q_{D3}}{L+1} \right)^2. \tag{19}$$

This is the first hard closure of one 10D diagnostic sector: backgrounds with similar low-energy vacuum energy but different tadpole mismatch no longer receive the same structural ranking.

For each candidate, the script also assigns simple dimensionless proxies for equation residuals, controlled flux activity, flux-sector alignment, robustness, compactification volume, and dimensional-control defects. These proxies enter the 10D support score of Eq. (4) and the structural penalty

$$\mathcal{F}_{\text{sel}}^{(10)}[\mathcal{B}] = -\log(\epsilon + \mathcal{M}_{10}^{\text{MAAT}}[\mathcal{B}]).$$

The implementation is intentionally transparent and reproducible:

`structural_selection_10d_tadpole_toy.py`

It produces the numerical data file

`structural_selection_10d_tadpole_toy_results.json`

and all figures in

`structural_selection_10d_tadpole_toy_plots/`.

11.1 Numerical Result

The default scan uses 320 candidates with random seed 4081. The key result is that energy proximity and tadpole mismatch are almost independent, while the 10D structural score does respond to tadpole closure:

$$r_s(|\Lambda_{\text{eff}}|, |\Delta Q_{D3}|) = 0.081, \qquad r_s(\mathcal{F}_{\text{sel}}^{(10)}, |\Delta Q_{D3}|) = 0.298.$$

Thus ordinary energy ranking does not see the charge-closure defect, but structural ranking does.

The top-20 candidates by energy proximity have mean

$$\langle |\Delta Q_{D3}| \rangle_{\text{energy}} = 4.40,$$

whereas the top-20 candidates by structural selection have

$$\langle |\Delta Q_{D3}| \rangle_{\text{struct}} = 1.90.$$

The combined phenomenological score improves this further to

$$\langle |\Delta Q_{D3}| \rangle_{\text{pheno}} = 1.35.$$

The overlap between the top-20 energy list and the top-20 structural list is only 6 candidates. This is the desired nontriviality condition: structural selection is not merely reproducing energy ranking.

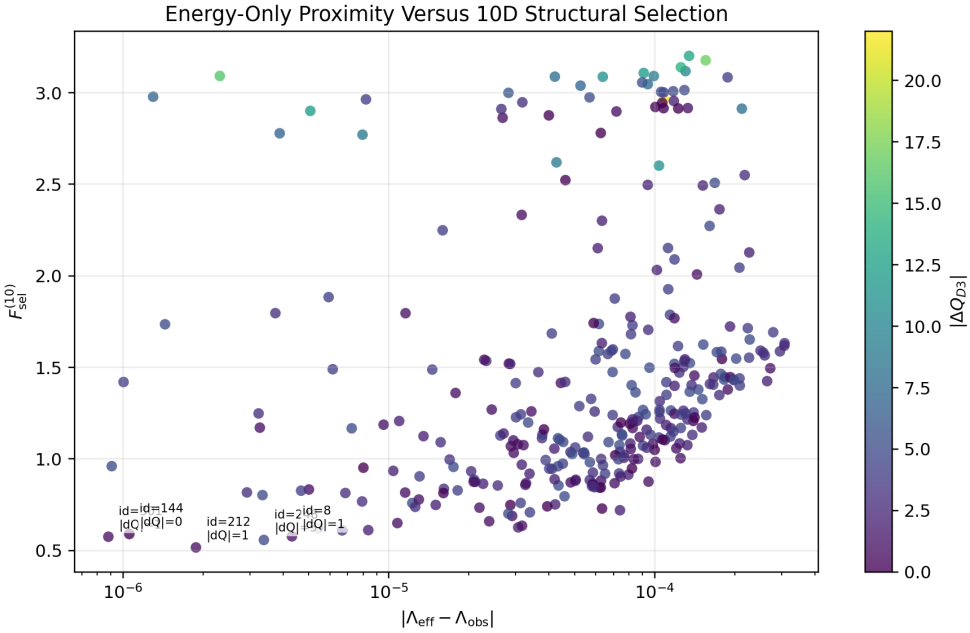

Figure 1: Energy proximity versus the 10D structural selection score. Colour denotes the absolute D3-tadpole residual. Candidates with similar $|\Lambda_{\rm eff}|$ can have very different structural penalties.

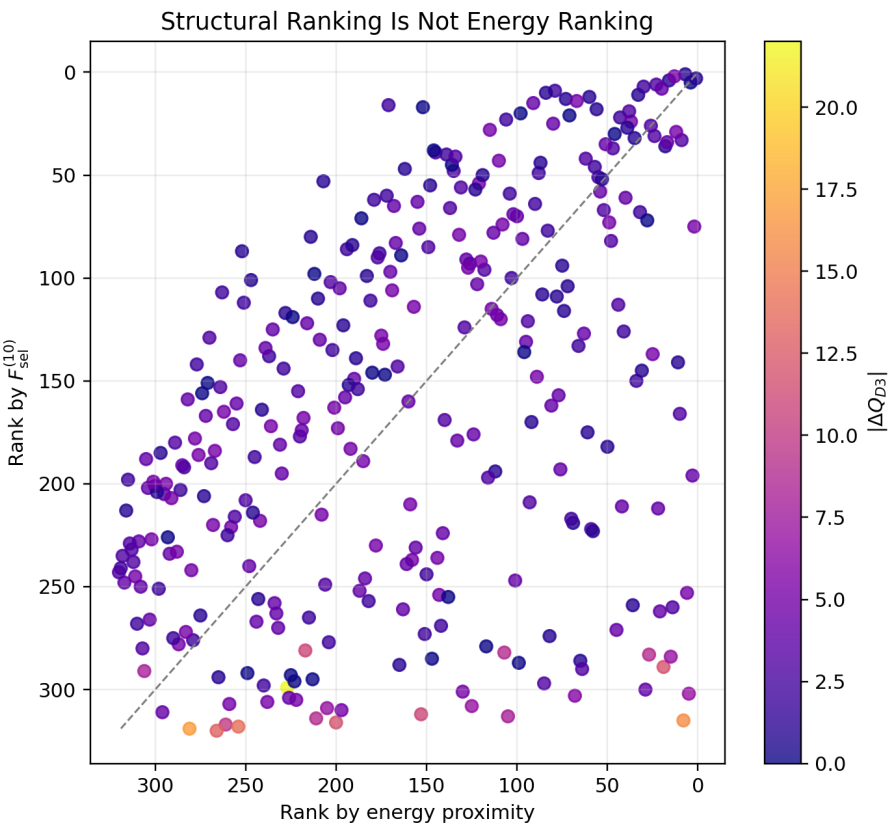

Figure 2: Rank comparison between energy-only ordering and structural ordering. The off-diagonal spread shows that $\mathcal{F}_{\rm sel}^{(10)}$ adds information beyond vacuum-energy proximity.

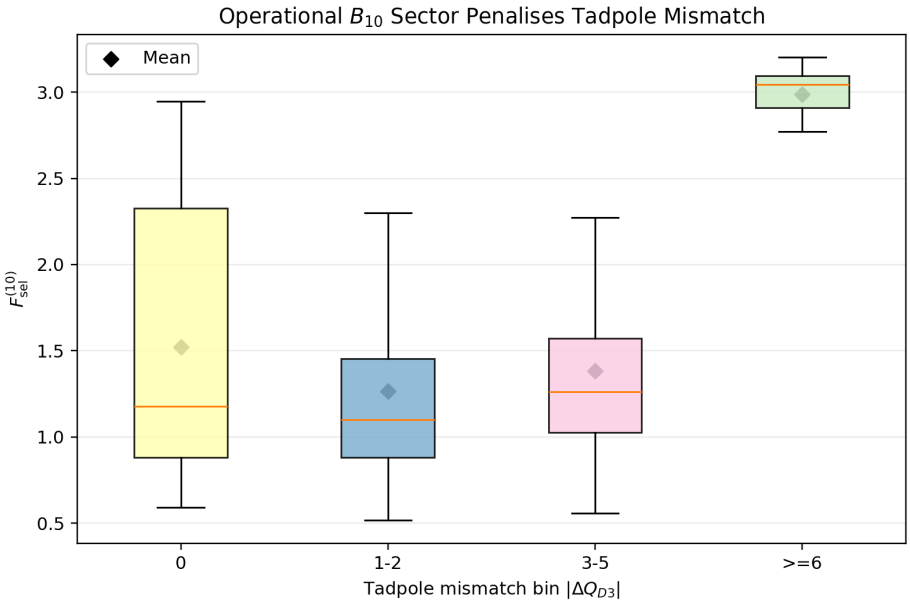

Figure 3: Distribution of $\mathcal{F}_{\rm sel}^{(10)}$ by tadpole-mismatch bin. The operational B_{10} sector penalises increasing charge mismatch.

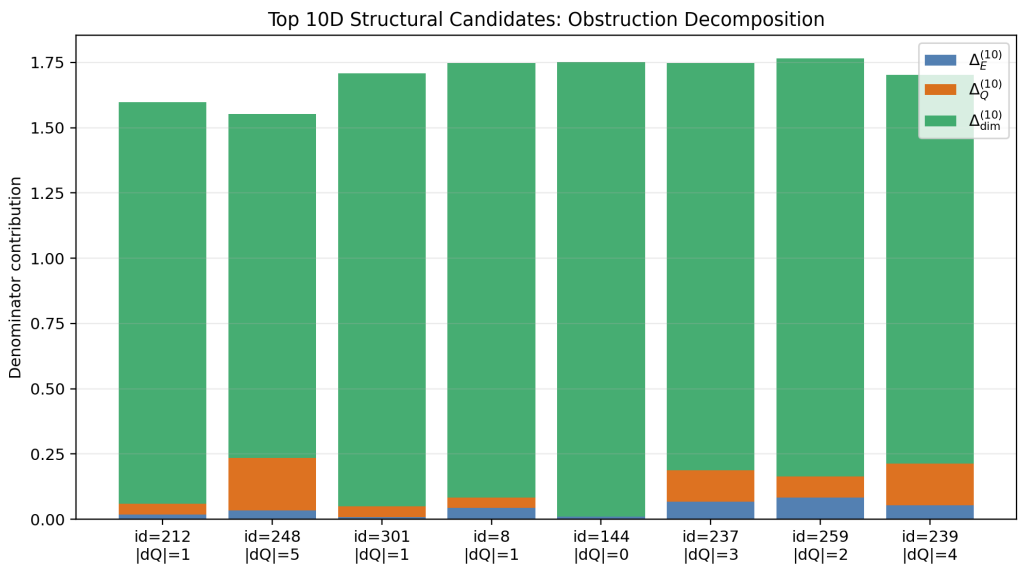

Figure 4: Obstruction decomposition for the top structural candidates. The test separates energy cost, tadpole cost, and dimensional-control cost in the 10D MAAT denominator.

11.2 Interpretation

This toy ensemble does not solve the full compactification problem. Its value is more modest but important: it demonstrates that the symbolic MAAT idea can be converted into a concrete 10D-near diagnostic with measurable consequences. The charge-closure sector becomes operational, the resulting ranking differs from energy-only selection, and the combined phenomenological measure selects backgrounds that are simultaneously closer to the observed vacuum energy and better aligned with tadpole closure.

In this sense, the first missing piece is partially addressed:

$$\text{abstract } B_{10} \quad \longrightarrow \quad \text{computable tadpole residual.}$$

The next step is to replace the remaining proxy sectors by compactification data from explicit type IIB, heterotic, or M-theory ensembles.

12 Second Minimal Test: A Flux-to-KKLT Bridge Ensemble

The previous test operationalises the B_{10} sector but still uses a toy vacuum-energy proxy. The next step is to connect the same flux/tadpole data to an actual reduced KKLT potential. We therefore construct a bridge ensemble in which integer flux vectors determine both a D3-tadpole residual and a small effective flux superpotential W_0.

The bridge pipeline is:

$$(F, H) \longrightarrow (N_{\text{flux}}, \Delta Q_{D3}, W_0, D) \longrightarrow V_{\text{KKLT}}(\tau, \theta) \longrightarrow \left(\tau, V_{\text{eff}}, m_i^2, \mathcal{F}_{\text{sel}}^{(10)}\right).$$

The reduced KKLT potential is

$$K = -3\log(T + \bar{T}), \qquad W = W_0 + Ae^{-aT}, \qquad V_{\text{eff}} = V_F + \frac{D}{(T + \bar{T})^n}.$$

Stationary points are found numerically in the (τ, θ) plane. The bridge score then combines:

- the 10D-near tadpole defect $d_B^{(10)}$,
- the KKLT stationarity and Hessian data,
- the vacuum-energy distance $|V_{\text{eff}} - \Lambda_{\text{obs}}|$,
- the volume-control defect $(10/\tau)^2 + e^{-a\tau}$,
- and a tachyonic-instability defect when present.

This is still not a real Calabi–Yau compactification scan. In particular, W_0 is generated by a controlled flux-projection proxy rather than by integrating the holomorphic three-form over an explicit geometry. The point of the test is narrower: it shows that the structural measure can be coupled to a real reduced KKLT dynamical system rather than only to symbolic flux data.

The implementation is:

`structural_selection_iib_kklt_bridge.py`

It produces:

`structural_selection_iib_kklt_bridge_results.json`

and figures in:

`structural_selection_iib_kklt_bridge_plots/`.

12.1 Bridge Results

The default bridge scan uses 90 flux backgrounds and finds 164 KKLT stationary candidates:

$$45 \text{ AdS minima}, \qquad 32 \text{ dS minima}, \qquad 87 \text{ unstable stationary points}.$$

The rank correlations are:

$$r_s(|V_{\text{eff}}|, \mathcal{F}^{(10)}_{\text{bridge}}) = 0.383,$$

$$r_s(|\Delta Q_{D3}|, |V_{\text{eff}}|) = 0.208, \qquad r_s(|\Delta Q_{D3}|, \mathcal{F}^{(10)}_{\text{bridge}}) = 0.566.$$

Thus the bridge score is more sensitive to tadpole closure than energy proximity alone.

The top-12 candidates by energy proximity have

$$\langle |\Delta Q_{D3}| \rangle_{\text{energy}} = 3.17, \qquad \langle \tau \rangle_{\text{energy}} = 106.76.$$

The top-12 candidates by bridge selection have

$$\langle |\Delta Q_{D3}| \rangle_{\text{bridge}} = 0.42, \qquad \text{median}(|\Delta Q_{D3}|) = 0, \qquad \langle \tau \rangle_{\text{bridge}} = 106.01.$$

Only 2 candidates overlap between the energy-top-12 and bridge-top-12 sets. The best energy candidate has rank 1 by energy but rank 22 by bridge selection, while the best bridge candidate has exact tadpole closure and rank 11 by energy.

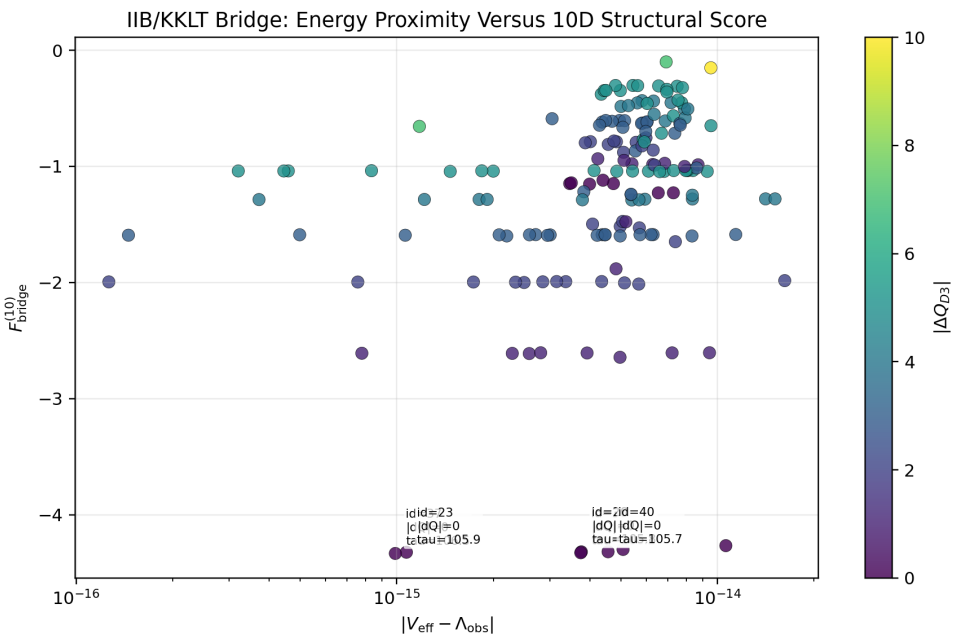

Figure 5: IIB/KKLT bridge ensemble: energy proximity versus the bridge structural score. Colour denotes the absolute D3-tadpole residual.

12.2 Interpretation

This bridge test addresses the main weakness of the pure toy model. The selection score is no longer evaluated on artificial vacuum-energy data alone; it is evaluated on actual stationary points of a reduced KKLT potential. The result remains phenomenological, but the hierarchy of tests is now clearer:

$$\text{symbolic 10D formula} \;\rightarrow\; \text{flux/tadpole toy ensemble} \;\rightarrow\; \text{flux-to-KKLT bridge ensemble}.$$

The next unresolved step is replacing the flux-projection proxy for W_0 by explicit period data from a concrete compactification geometry.

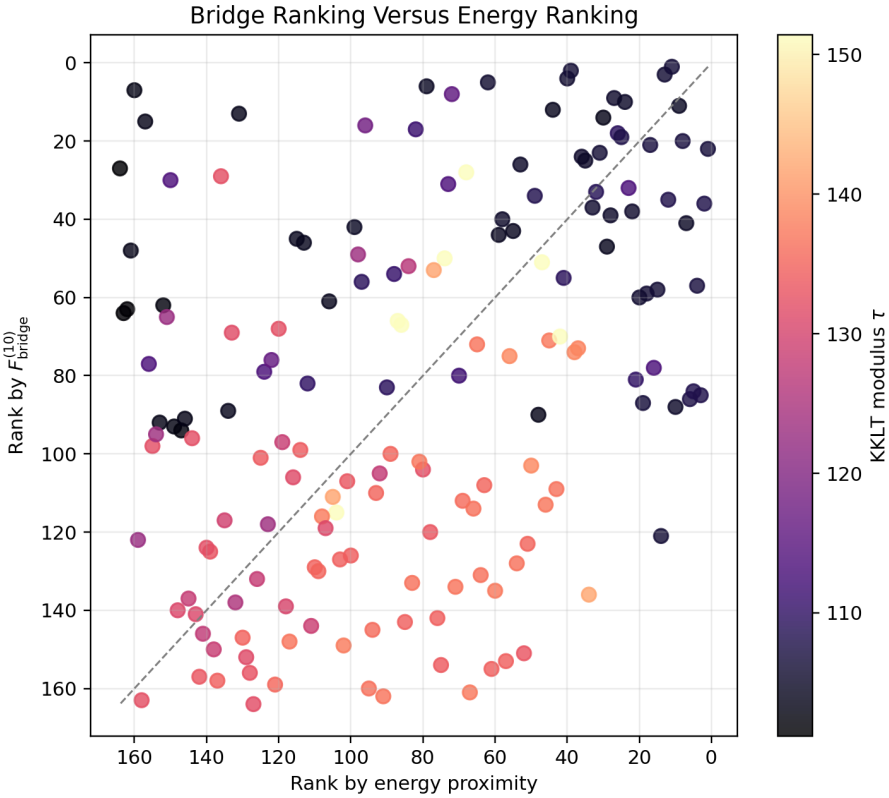

Figure 6: Rank comparison between energy-only ordering and bridge structural ordering. The bridge score selects a different top region of the ensemble.

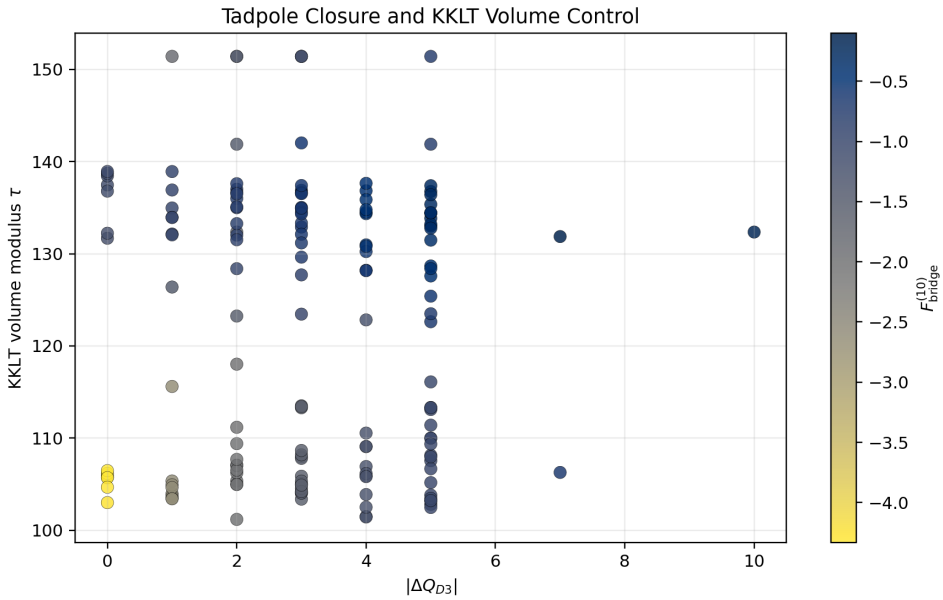

Figure 7: Tadpole residual versus KKLT volume modulus τ, coloured by bridge structural score. The best region combines charge closure with controlled large-volume stationarity.

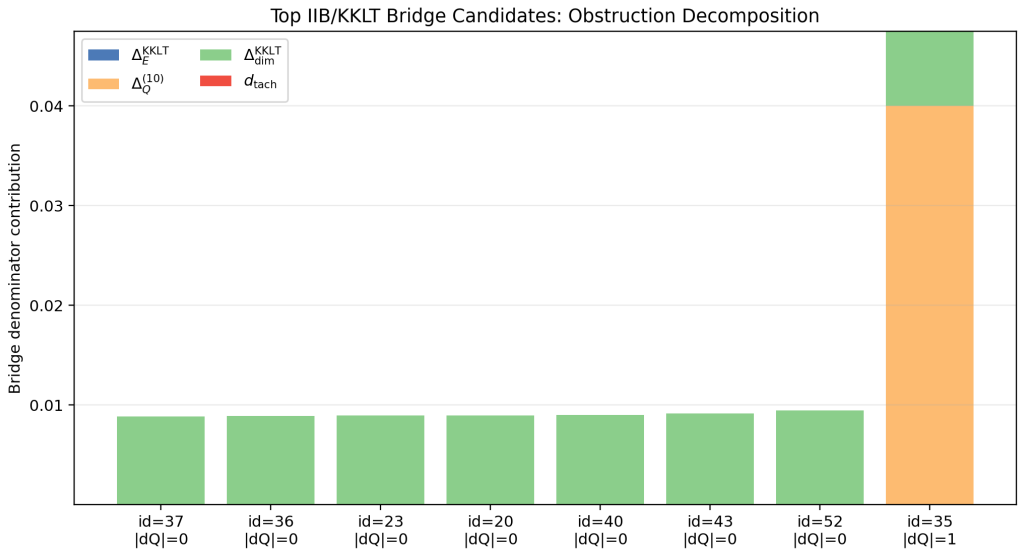

Figure 8: Obstruction decomposition for the top bridge candidates. The score combines KKLT energy distance, 10D tadpole mismatch, dimensional control, and tachyonic instability penalties.

13 Third Minimal Test: Period Data to KKLT

The bridge test above still maps fluxes to W_0 through a controlled proxy. The next improvement is to compute the flux superpotential from an explicit period vector. We therefore introduce a one-parameter large-complex-structure benchmark. It is inspired by the mirror-quintic period structure, but should be understood as a controlled period-data test rather than a full instanton-corrected compactification database.

Let $t = x + iy$ be a complex-structure modulus in the large-complex-structure regime. We use the period vector

$$\Pi_{\mathrm{LCS}}(t) = \left(1,\, t,\, \frac{\kappa}{2}t^2 + \frac{c_2\cdot J}{24},\, -\frac{\kappa}{6}t^3 - \frac{c_2\cdot J}{24}t + i\,\frac{\chi\,\zeta(3)}{2(2\pi)^3}\right), \tag{20}$$

with benchmark values

$$\kappa = 5, \qquad c_2\cdot J = 50, \qquad \chi = -200.$$

Given integer flux vectors $F, H \in \mathbb{Z}^4$ and an axio-dilaton $\tau_{\mathrm{ax}} = C_0 + i/g_s$, the raw flux superpotential is

$$W_{0,\mathrm{raw}} = (F - \tau_{\mathrm{ax}} H)\cdot \Pi_{\mathrm{LCS}}(t). \tag{21}$$

Its magnitude is then mapped into the small real W_0 range used by the reduced KKLT solver. This last rescaling is a benchmark normalisation, not a claim that the exact compactification measure has been derived.

The full pipeline is now

$$(F, H, t, \tau_{\mathrm{ax}}) \longrightarrow (W_{0,\mathrm{raw}}, W_0, \Delta Q_{D3}) \longrightarrow V_{\mathrm{KKLT}}(\tau,\theta) \longrightarrow \mathcal{F}^{(10)}_{\mathrm{period}}.$$

The script implementing this test is:

`structural_selection_iib_period_kklt_bridge.py`

It generates:

`structural_selection_iib_period_kklt_bridge_results.json`

and figures in:

`structural_selection_iib_period_kklt_bridge_plots/`.

13.1 Period-Bridge Results

The default period-to-KKLT scan uses 120 period backgrounds and finds 224 KKLT stationary candidates:

Candidate type	**Count**
AdS minima	60
dS minima	42
near-Minkowski minima	5
unstable stationary points	117

The rank correlations are:

$$r_s(|V_{\text{eff}}|, \mathcal{F}^{(10)}_{\text{period}}) = -0.034,$$

$$r_s(|\Delta Q_{D3}|, |V_{\text{eff}}|) = 0.040, \qquad r_s(|\Delta Q_{D3}|, \mathcal{F}^{(10)}_{\text{period}}) = 0.582.$$

Thus, after replacing the direct W_0 proxy by a period calculation, energy proximity remains almost blind to tadpole closure, while the structural score remains sensitive to it.

The top-12 candidates by energy proximity have

$$\langle|\Delta Q_{D3}|\rangle_{\text{energy}} = 2.83, \qquad \langle|V_{\text{eff}}|\rangle_{\text{energy}} = 2.85 \times 10^{-16}.$$

The top-12 candidates by period-structural score have

$$\langle|\Delta Q_{D3}|\rangle_{\text{period}} = 0.67, \qquad \langle\text{Im}(t)\rangle_{\text{period}} = 3.35,$$

but a larger mean energy distance,

$$\langle|V_{\text{eff}}|\rangle_{\text{period}} = 9.02 \times 10^{-15}.$$

This is not a failure; it is the intended separation between structural prior and observational conditioning. When the phenomenological likelihood is included, the top-12 candidates satisfy

$$\langle|\Delta Q_{D3}|\rangle_{\text{pheno}} = 0.83, \qquad \langle|V_{\text{eff}}|\rangle_{\text{pheno}} = 2.20 \times 10^{-15}.$$

The best phenomenological candidate has exact tadpole closure, energy rank 8, period-structural rank 6, and phenomenological rank 1. This is the desired behaviour: the combined measure does not merely choose the lowest energy candidate or the most structurally ideal candidate, but a compromise between both.

13.2 Interpretation

This third test closes the next gap in the chain. The effective W_0 is no longer inserted directly, but computed from fluxes, an axio-dilaton, and a large-complex-structure period vector. The construction remains approximate: it omits instanton corrections, uses a simplified period basis, and rescales $|W_{0,\text{raw}}|$ into the KKLT range. Nevertheless, it establishes a concrete route from geometric period data to structural vacuum ranking:

$$\text{period data} \;\rightarrow\; W_0 \;\rightarrow\; \text{KKLT stationary point} \;\rightarrow\; \text{structural selection}.$$

The next step is therefore sharply defined: replace the LCS benchmark period vector by exact periods from an explicit compactification geometry.

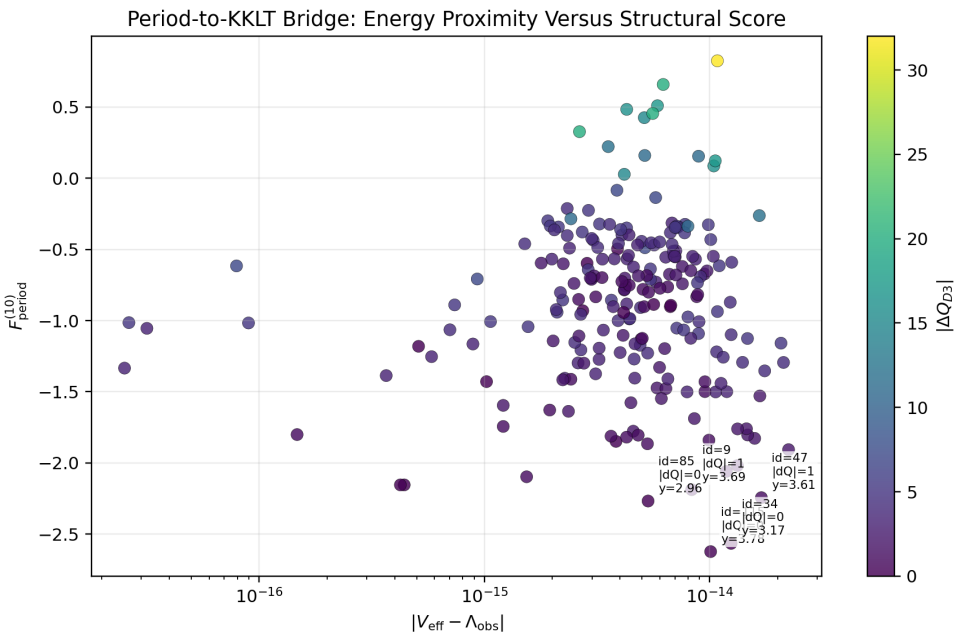

Figure 9: Period-to-KKLT bridge ensemble: energy proximity versus the period structural score. Colour denotes the absolute D3-tadpole residual.

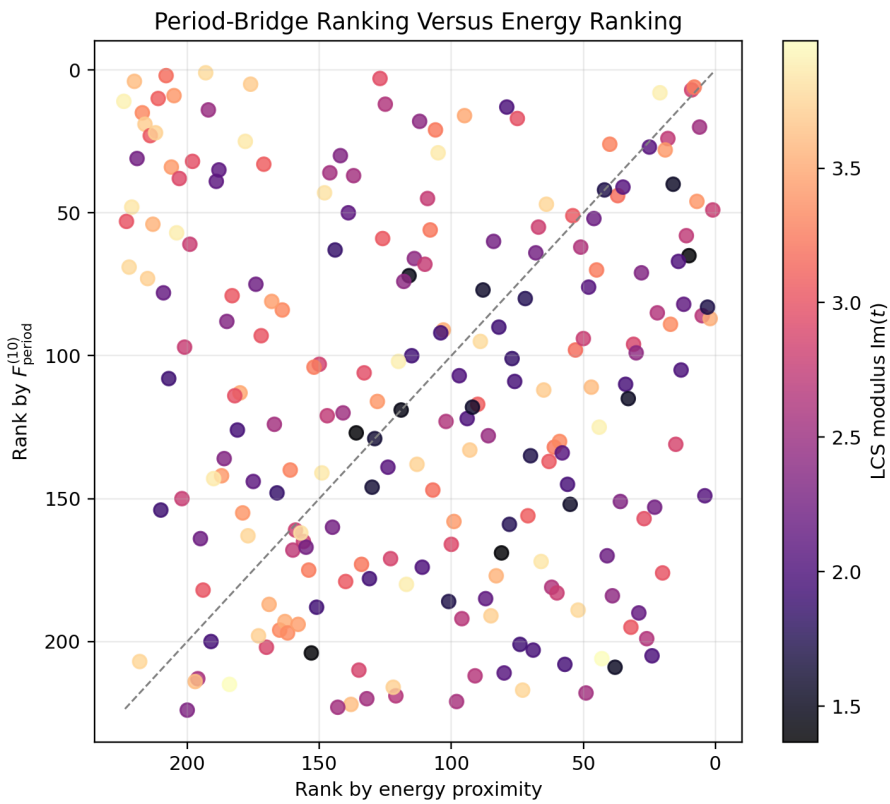

Figure 10: Rank comparison between energy-only ordering and period-structural ordering. The period score selects a different region of the ensemble.

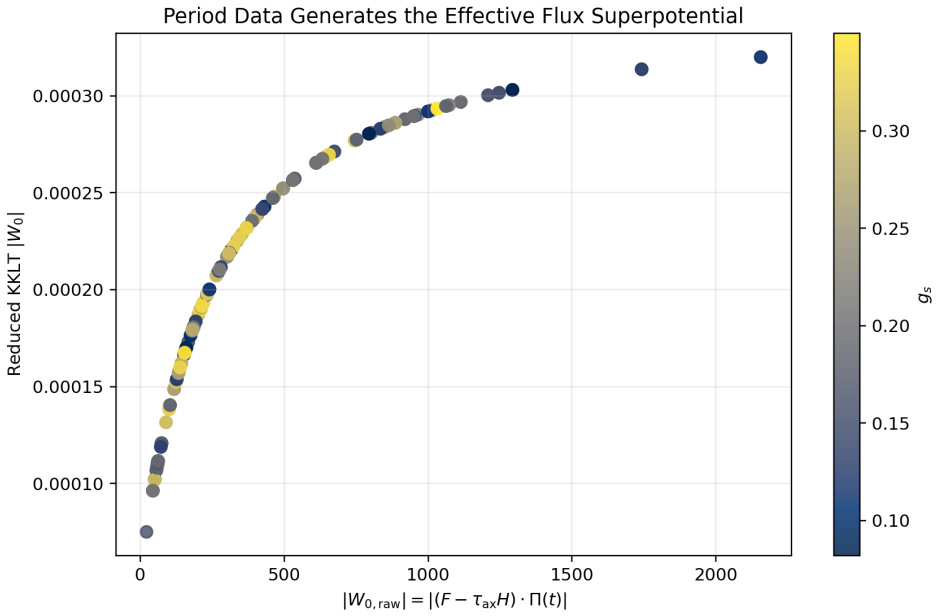

Figure 11: The period calculation maps raw flux-period data $|(F - \tau_{\mathrm{ax}} H) \cdot \Pi(t)|$ into the reduced KKLT superpotential $|W_0|$. The rescaling is a benchmark normalisation.

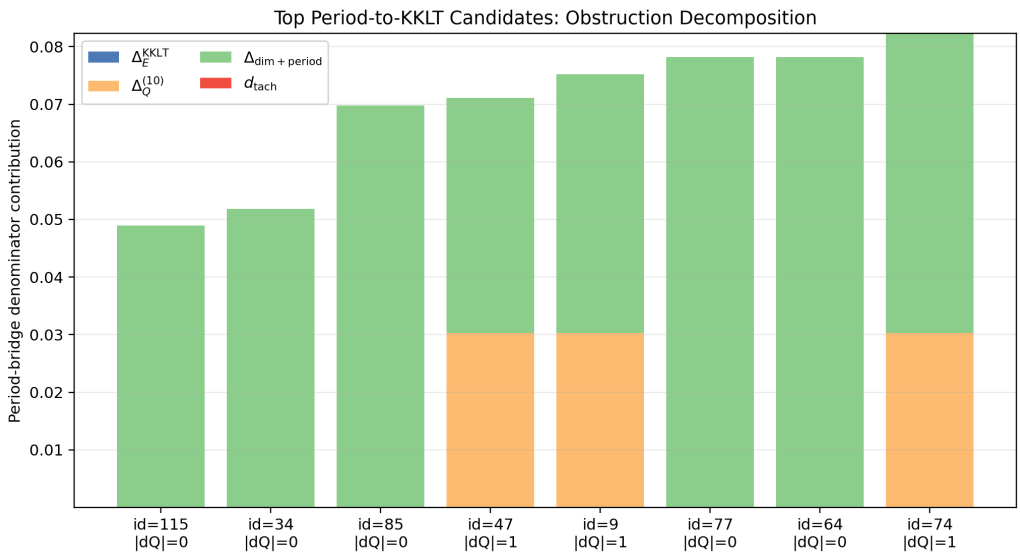

Figure 12: Obstruction decomposition for the top period-to-KKLT candidates. The denominator combines KKLT energy distance, tadpole mismatch, period/dimensional control, and tachyonic instability penalties.

14 Fourth Minimal Test: Picard–Fuchs Periods of the Mirror Quintic

The previous section still used a compact LCS period model. We now replace this by numerical periods of a concrete one-parameter geometry: the mirror quintic. The Picard–Fuchs operator is

$$\mathcal{L}_{\mathrm{PF}} = \theta^4 - 5z(5\theta+1)(5\theta+2)(5\theta+3)(5\theta+4), \qquad \theta = z\frac{\mathrm{d}}{\mathrm{d}z}. \tag{22}$$

Near the large-complex-structure point $z = 0$, the fundamental series is

$$\omega_0(z) = \sum_{n=0}^{N} \frac{(5n)!}{(n!)^5} z^n, \tag{23}$$

and the logarithmic periods are obtained by the Frobenius construction. In the implementation, a generating solution

$$\omega(\rho, z) = \sum_{n=0}^{N} \frac{\Gamma(5(n+\rho)+1)}{\Gamma(n+\rho+1)^5} z^{n+\rho}$$

is differentiated with respect to ρ at $\rho = 0$ to obtain four local periods

$$\Pi_{\mathrm{PF}}(z) = (\omega_0, \omega_1, \omega_2, \omega_3)\,.$$

The truncation used in the default benchmark is $N = 42$ terms at 50 decimal digits of precision. Candidates are sampled within

$$10^{-6} \le |z| \le 1.8 \times 10^{-4},$$

inside the convergence region of the large-complex-structure expansion.

The flux superpotential is now computed from the Picard–Fuchs period vector:

$$W_{0,\mathrm{raw}} = (F - \tau_{\mathrm{ax}} H) \cdot \Pi_{\mathrm{PF}}(z). \tag{24}$$

As before, $|W_{0,\mathrm{raw}}|$ is mapped into the small real W_0 interval used by the reduced KKLT solver. This rescaling is a benchmark convention; the new point is that the raw value itself is generated by a concrete Picard–Fuchs period calculation.

The implementation is:

`structural_selection_iib_exact_period_kklt_bridge.py`

It generates:

`structural_selection_iib_exact_period_kklt_bridge_results.json`

and figures in:

`structural_selection_iib_exact_period_kklt_bridge_plots/`.

14.1 Exact-Period Results

The default exact-period scan uses 110 mirror-quintic period backgrounds and finds 219 KKLT stationary candidates:

Candidate type	**Count**
dS minima	57
AdS minima	43
near-Minkowski minima	1
unstable stationary points	118

The rank correlations are:

$$r_s(|V_{\text{eff}}|, \mathcal{F}^{(10)}_{\text{exact}}) = 0.060,$$

$$r_s(|\Delta Q_{D3}|, |V_{\text{eff}}|) = 0.072, \qquad r_s(|\Delta Q_{D3}|, \mathcal{F}^{(10)}_{\text{exact}}) = 0.582.$$

The structural score remains strongly sensitive to tadpole closure, while energy proximity remains nearly blind to it.

The top-12 candidates by energy proximity have

$$\langle|\Delta Q_{D3}|\rangle_{\text{energy}} = 4.67, \qquad \langle|V_{\text{eff}}|\rangle_{\text{energy}} = 5.71 \times 10^{-16}.$$

The top-12 candidates by exact-period structural score have

$$\langle|\Delta Q_{D3}|\rangle_{\text{exact}} = 0.50, \qquad \langle|V_{\text{eff}}|\rangle_{\text{exact}} = 5.51 \times 10^{-15}.$$

The phenomenological score, which includes the vacuum-energy likelihood, gives

$$\langle|\Delta Q_{D3}|\rangle_{\text{pheno}} = 0.75, \qquad \langle|V_{\text{eff}}|\rangle_{\text{pheno}} = 2.15 \times 10^{-15}.$$

The top-12 energy list and top-12 exact-period structural list have zero overlap. This is the strongest nontriviality result in the current chain: energy ordering and exact-period structural ordering select genuinely different regions of the ensemble.

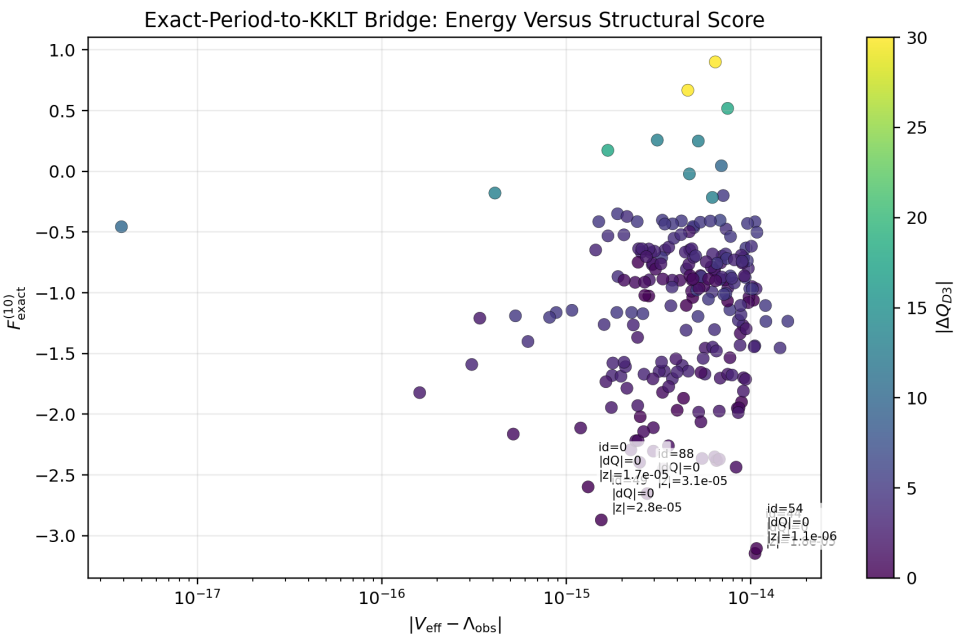

Figure 13: Mirror-quintic Picard–Fuchs period bridge: energy proximity versus the exact-period structural score. Colour denotes the absolute D3-tadpole residual.

14.2 Interpretation

This fourth test closes the next conceptual gap. The background data now pass through an explicit Picard–Fuchs period calculation for a concrete one-parameter geometry before entering the KKLT and structural-selection pipeline. The result is still not a full landscape scan: the period basis is local, the series is truncated, instanton corrections away from the LCS patch are not globally continued, and the conversion from $|W_{0,\text{raw}}|$ to the reduced KKLT scale is a benchmark normalisation.

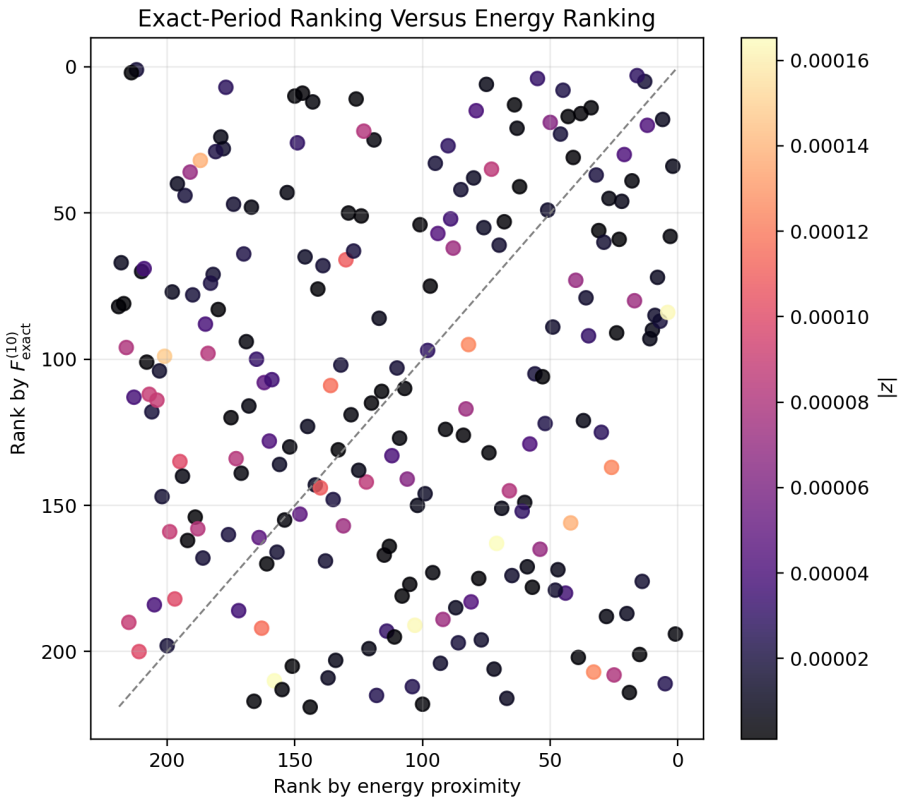

Figure 14: Rank comparison between energy-only ordering and exact-period structural ordering. The top lists have zero overlap at the top-12 level.

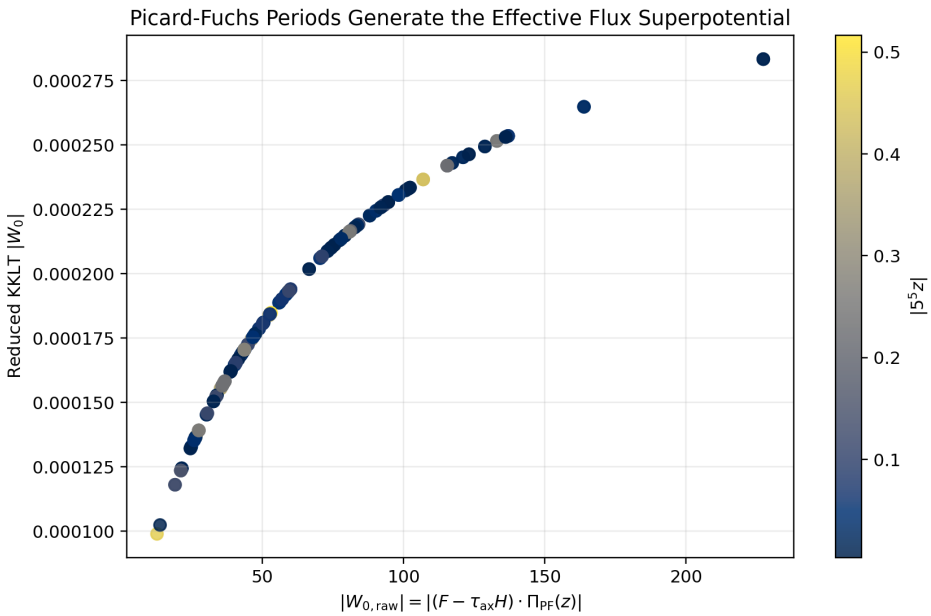

Figure 15: The Picard–Fuchs period calculation maps $|(F - \tau_{\text{ax}}H) \cdot \Pi_{\text{PF}}(z)|$ into the reduced KKLT superpotential $|W_0|$.

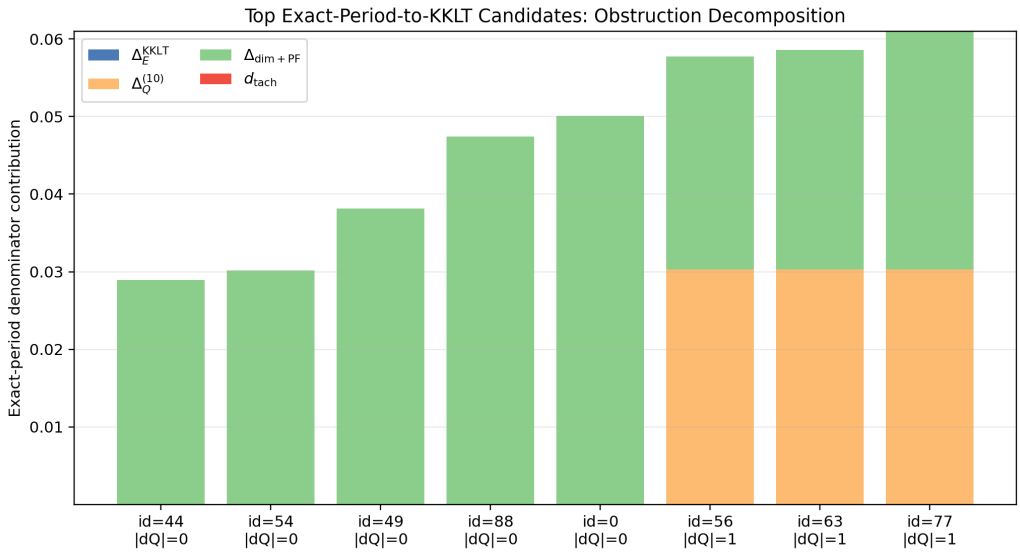

Figure 16: Obstruction decomposition for the top exact-period-to-KKLT candidates. The denominator combines KKLT energy distance, tadpole mismatch, exact-period control, and tachyonic instability penalties.

Nevertheless, the chain has advanced from symbolic formula to concrete geometry:

$$\text{Picard–Fuchs periods} \;\to\; W_0 \;\to\; \text{KKLT stationary points} \;\to\; \text{structural ranking.}$$

The remaining hard problem is no longer whether such a pipeline can be written; it is whether the same ranking effect persists in exact global period data and larger compactification ensembles.

15 Fifth Minimal Test: Backreaction and Standard-Model Sector Proxies

The exact-period benchmark still omits two physically important layers: ten-dimensional backreaction and Standard-Model-like low-energy structure. A complete treatment would require solving the full ten-dimensional equations with localized sources, orientifolds, warping, Bianchi identities, and a realistic brane or bundle sector. That is beyond the scope of the present phenomenological paper. However, both missing layers can be introduced as operational diagnostic proxies.

15.1 Backreaction Proxy

In type IIB language, the warp factor is controlled schematically by a Poisson equation of the form

$$\Delta_6 e^{-4A} \sim \rho_{\mathrm{flux}} + \rho_{\mathrm{loc}} - \langle \rho_{\mathrm{flux}} + \rho_{\mathrm{loc}} \rangle. \tag{25}$$

The benchmark implements a discrete version of this idea on a small internal toroidal lattice. Local positive and negative source charges are placed on the grid, a mean-subtracted source density is formed, and a spectral Poisson solve produces a warp-potential proxy φ_A. The backreaction defect is then defined from the warp amplitude, gradient energy, source clumping, and net tadpole mismatch:

$$d_{\mathrm{back}} = c_1 \langle \varphi_A^2 \rangle + c_2 \langle |\nabla \varphi_A|^2 \rangle + c_3 \left(\frac{|\Delta Q_{D3}|}{L+1} \right)^2 + c_4 \,\mathrm{clump}(\rho_{\mathrm{loc}}). \tag{26}$$

This is not a 10D supergravity solution. It is a controlled diagnostic for whether localized source data are likely to induce stronger warping and backreaction.

15.2 Standard-Model Sector Proxy

The low-energy Standard-Model sector is represented by a separate likelihood. The benchmark assigns topological proxy data to each background:

$$\mathrm{rank}(G), \qquad N_{\mathrm{gen}}, \qquad m_Y, \qquad \mathrm{rank}(Y_{\mathrm{Yukawa}}),$$

interpreted respectively as gauge-rank, generation-count, hypercharge-mass, and Yukawa-rank diagnostics. The SM-sector defect is

$$d_{\mathrm{SM}} = \left(\frac{\mathrm{rank}(G) - 12}{6}\right)^2 + \left(\frac{N_{\mathrm{gen}} - 3}{3}\right)^2 + m_Y^2 + \left(\frac{3 - \mathrm{rank}(Y_{\mathrm{Yukawa}})}{3}\right)^2 + \cdots . \tag{27}$$

The corresponding likelihood is

$$\mathcal{L}_{\mathrm{SM}} = \exp[-d_{\mathrm{SM}}/\sigma_{\mathrm{SM}}]. \tag{28}$$

Again, this is not yet a genuine Standard-Model compactification. It is the first explicit low-energy selection layer that distinguishes structurally good vacua from vacua that also look more SM-like.

The full benchmark score is

$$\mathcal{F}_{\mathrm{full}}^{(10+\mathrm{SM})} = \mathcal{F}_{\mathrm{exact}}^{(10)} + w_{\mathrm{back}} d_{\mathrm{back}} + w_{\mathrm{SM}} d_{\mathrm{SM}} + \mathcal{F}_{\mathrm{SW}} + \mathcal{F}_{\Lambda}. \tag{29}$$

The implementation is:

`structural_selection_iib_backreaction_sm_bridge.py`

It generates:

`structural_selection_iib_backreaction_sm_bridge_results.json`

and figures in:

`structural_selection_iib_backreaction_sm_bridge_plots/`.

15.3 Backreaction–SM Results

The default scan uses 90 exact-period backgrounds and finds 172 KKLT stationary candidates:

Candidate type	**Count**
dS minima	47
AdS minima	34
near-Minkowski minima	1
unstable stationary points	90

The full score correlates with the new diagnostic axes as expected:

$$r_s(|V_{\mathrm{eff}}|, \mathcal{F}_{\mathrm{full}}) = 0.432, \qquad r_s(d_{\mathrm{SM}}, \mathcal{F}_{\mathrm{full}}) = 0.282,$$

$$r_s(d_{\mathrm{back}}, \mathcal{F}_{\mathrm{full}}) = 0.081, \qquad r_s(|\Delta Q_{D3}|, \mathcal{F}_{\mathrm{full}}) = 0.340.$$

The top-12 candidates by energy proximity have

$$\langle |\Delta Q_{D3}| \rangle_{\mathrm{energy}} = 3.33, \qquad \langle d_{\mathrm{back}} \rangle_{\mathrm{energy}} = 0.0466, \qquad \langle d_{\mathrm{SM}} \rangle_{\mathrm{energy}} = 1.485.$$

The top-12 candidates by the full backreaction+SM score have

$$\langle|\Delta Q_{D3}|\rangle_{\text{full}} = 0.92, \qquad \langle d_{\text{back}}\rangle_{\text{full}} = 0.0439, \qquad \langle d_{\text{SM}}\rangle_{\text{full}} = 0.886.$$

The top-12 overlap between energy ranking and full ranking is only one candidate. The best full candidate has exact tadpole closure, reduced backreaction defect, lower SM-sector defect, and energy rank 71 rather than rank 1. This demonstrates the purpose of the additional layers: the selected candidate need not be the energy-closest one if it is structurally and phenomenologically better controlled.

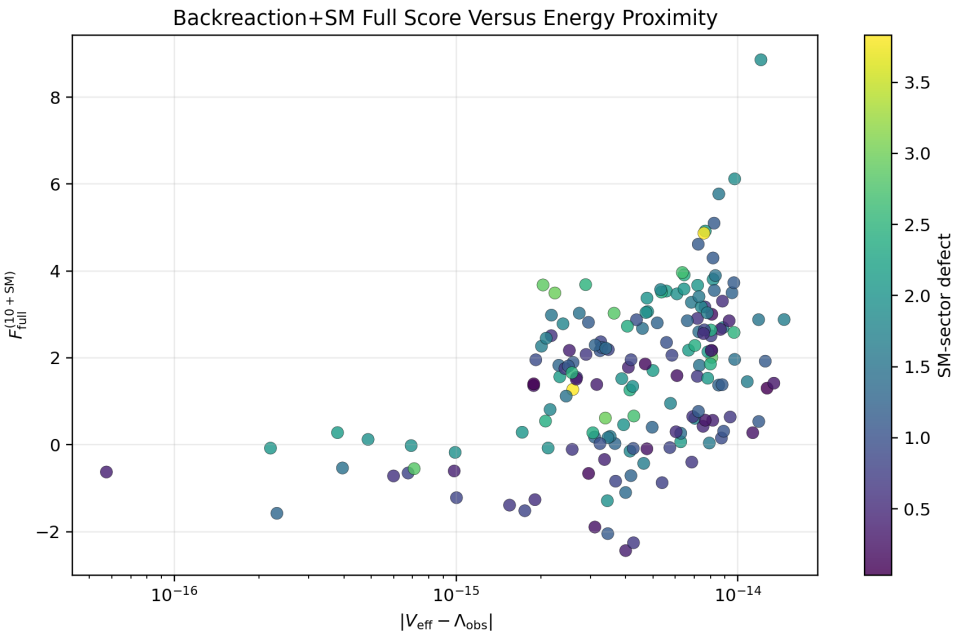

Figure 17: Full backreaction+SM score versus energy proximity. Colour denotes the SM-sector defect.

15.4 Interpretation

This fifth test does not solve the full backreaction problem and does not construct the Standard Model. It does something more modest but necessary: it gives both issues an explicit mathematical place in the selection measure. Backreaction enters as an operational warp/source defect; the Standard-Model sector enters as a likelihood-like low-energy diagnostic. The resulting score therefore implements the hierarchy

$$\text{geometry} \to \text{periods} \to \text{KKLT dynamics} \to \text{10D consistency proxy} \to \text{SM-sector proxy}.$$

The next hard step is replacing these proxies by full 10D source solutions. In parallel, the SM proxy must be replaced by explicit brane or bundle constructions with real gauge group and chiral spectrum.

16 Unified Phenomenological Master Functional

The preceding sections introduced the ingredients one layer at a time. We now collect them into a single phenomenological master formula. This is not claimed to be a final microscopic Theory of Everything. It is a defined effective selection functional that can be evaluated once a regulated ensemble of string backgrounds is specified.

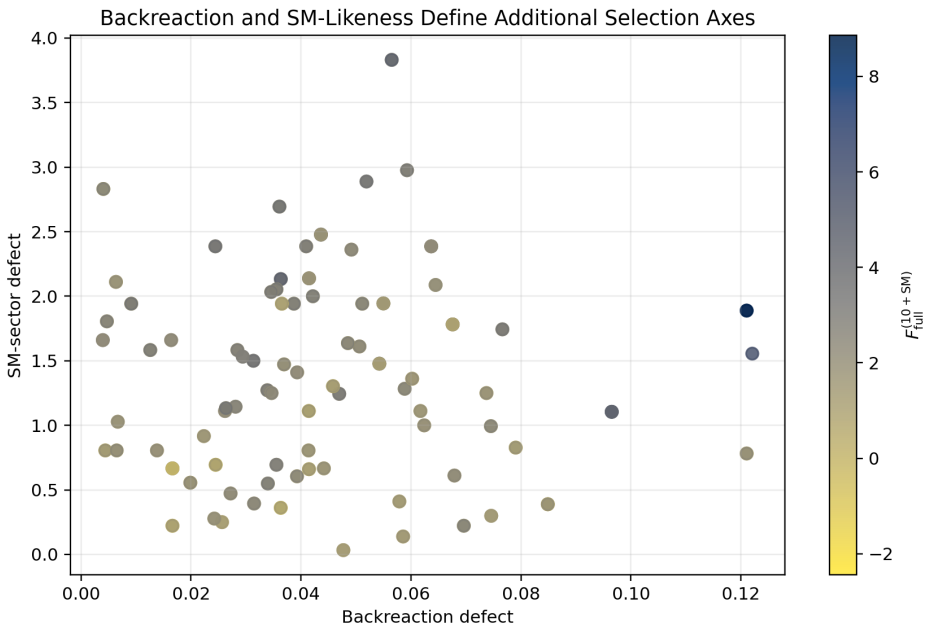

Figure 18: Backreaction defect versus SM-sector defect, coloured by the full phenomenological score. The new axes separate vacua that were indistinguishable under energy or period data alone.

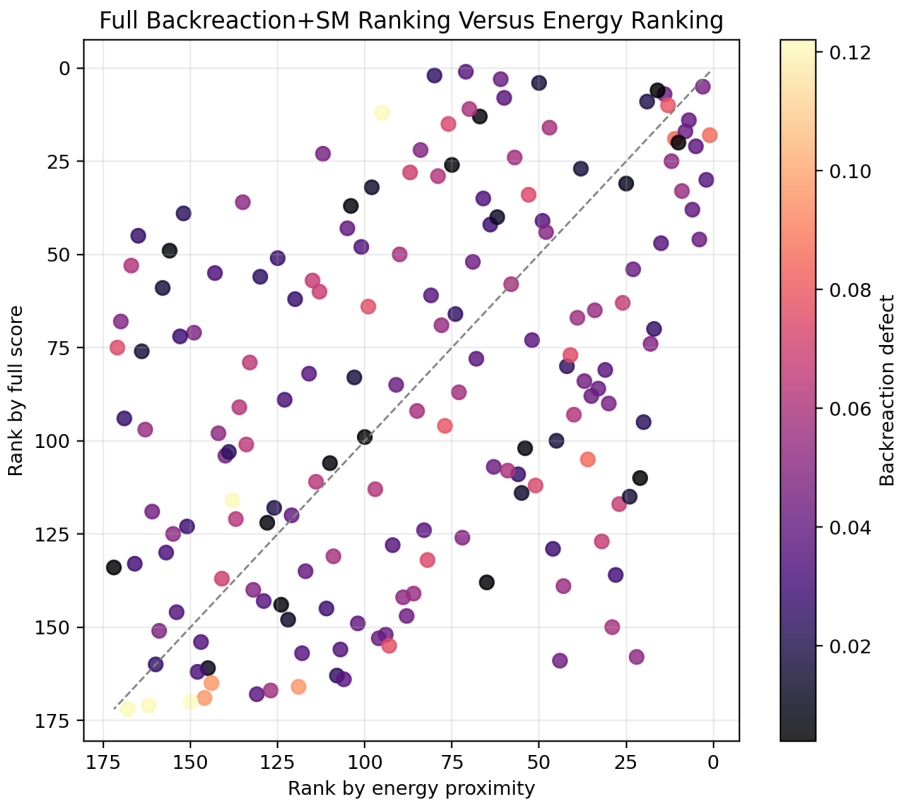

Figure 19: Rank comparison between energy-only ordering and full backreaction+SM ordering. The near-diagonal energy ranking is strongly broken.

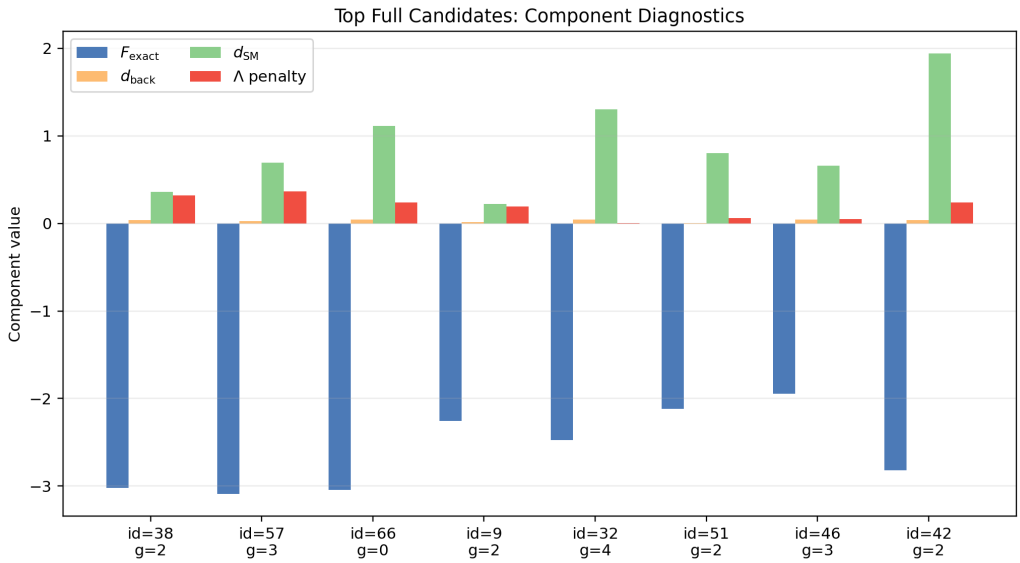

Figure 20: Component diagnostics for the top full candidates. The full score is not a pure energy score; it combines exact-period structure, backreaction, SM-sector quality, and vacuum-energy conditioning.

Let a candidate background be

$$\mathcal{B} = (\mathcal{M}_{10}, G_{10}, \Phi, H_3, F_p, \mathcal{S}_{\text{loc}}, z, \tau_{\text{ax}}, T, \mathcal{G}_{\text{SM}}) \,,$$

where $\mathcal{S}_{\text{loc}}$ denotes localized sources such as branes and orientifolds, z denotes complex-structure data, τ_{ax} the axio-dilaton, T a Kähler modulus, and $\mathcal{G}_{\text{SM}}$ the low-energy gauge/matter sector data.

The unified selected measure is

$$\boxed{\mathrm{d}\mathbb{P}_{\text{unified}}(\mathcal{B} \mid \mathcal{D}) = \frac{1}{Z_{\mathcal{D}}}\, \chi_{\text{adm}}[\mathcal{B}]\, \exp[-\mathcal{F}_{\text{unified}}[\mathcal{B};\mathcal{D}]]\, \mathrm{d}\mu_{\text{scan}}[\mathcal{B}].} \tag{30}$$

The corresponding master functional is

$$\boxed{\begin{aligned}\mathcal{F}_{\text{unified}}[\mathcal{B};\mathcal{D}] = \mathcal{F}_{\text{geom}}[\mathcal{B}] + \mathcal{F}_{\text{period}}[\mathcal{B}] + \mathcal{F}_{\text{KKLT}}[\mathcal{B}] + \mathcal{F}_{\text{back}}[\mathcal{B}] + \mathcal{F}_{\text{SM}}[\mathcal{B}] \\ + \mathcal{F}_{\text{SW}}[\mathcal{B}] - \log W_{\cos}[\mathcal{B}] - \log \mathcal{L}_{\text{obs}}(\mathcal{D} \mid \mathcal{B}).\end{aligned}} \tag{31}$$

Hard inconsistency is handled by $\chi_{\text{adm}}[\mathcal{B}] = 0$, which is equivalent to an infinite penalty.

16.1 Geometric and Structural Core

The geometric structural core is the 10D MAAT support already introduced in Eq. (4):

$$\mathcal{F}_{\text{geom}}[\mathcal{B}] = -\log\big(\epsilon + \mathcal{M}_{10}^{\text{MAAT}}[\mathcal{B}]\big) \,. \tag{32}$$

Equivalently, in defect-sector notation,

$$\mathcal{F}_{\text{geom}}[\mathcal{B}] = -\sum_{a\in\{H,B,S,V,R,\Sigma\}} \lambda_a \log(\epsilon + \Gamma_a[\mathcal{B}]), \qquad \Gamma_a = \frac{1}{1+d_a}. \tag{33}$$

This term contains equation closure, constraint closure, controlled activity, inter-sector coupling, robustness, and global compactification integrity.

16.2 Period Sector

The period sector penalises loss of control in the complex-structure and flux superpotential computation. In the strongest benchmark of this paper,

$$W_{0,\mathrm{raw}} = (F - \tau_{\mathrm{ax}} H) \cdot \Pi_{\mathrm{PF}}(z), \tag{34}$$

where Π_{PF} is obtained from the mirror-quintic Picard–Fuchs system. A minimal period-control penalty is

$$\mathcal{F}_{\mathrm{period}}[\mathcal{B}] = \lambda_{\mathrm{per}} \left[|5^5 z|^2 + g_s^2 + d_{\mathrm{basis}} + d_{\mathrm{trunc}} \right]. \tag{35}$$

Here d_{basis} marks whether the period basis is merely local or globally symplectic, while d_{trunc} controls numerical truncation. In the present benchmark these are effective proxies; in a full compactification scan they should be replaced by exact global-period data.

16.3 KKLT and Vacuum-Dynamics Sector

The reduced KKLT sector is

$$\mathcal{F}_{\mathrm{KKLT}}[\mathcal{B}] = \lambda_E \frac{|V_{\mathrm{eff}} - \Lambda_{\mathrm{obs}}|}{V_{\mathrm{scale}}} + \lambda_{\mathrm{stat}} \|\nabla V_{\mathrm{eff}}\|^2 + \lambda_{\mathrm{tach}} d_{\mathrm{tach}} + \lambda_{\mathrm{EFT}} d_{\mathrm{EFT}}. \tag{36}$$

This is the bridge between period data and four-dimensional vacuum structure. It measures whether the candidate produces a controlled stationary point with acceptable EFT behaviour and stability margins.

16.4 Backreaction and Local-Source Sector

The backreaction term is the first placeholder for full ten-dimensional localized-source physics:

$$\mathcal{F}_{\mathrm{back}}[\mathcal{B}] = \lambda_{\mathrm{warp}} \langle \varphi_A^2 \rangle + \lambda_{\nabla A} \langle |\nabla \varphi_A|^2 \rangle + \lambda_{\mathrm{src}} d_{\mathrm{clump}} + \lambda_Q \left(\frac{|\Delta Q_{D3}|}{L+1} \right)^2. \tag{37}$$

In a complete theory, φ_A should be replaced by the actual warp factor obtained from the ten-dimensional equations. In the benchmark it is computed by a discrete Poisson solve and is therefore a backreaction proxy.

16.5 Standard-Model Sector

The Standard-Model sector is included through the negative log-likelihood:

$$\mathcal{F}_{\mathrm{SM}}[\mathcal{B}] = -\log \mathcal{L}_{\mathrm{SM}}(D_{\mathrm{SM}} \mid \mathcal{B}). \tag{38}$$

At proxy level,

$$\begin{aligned} \mathcal{F}_{\mathrm{SM}}[\mathcal{B}] = \lambda_{\mathrm{rank}} \left(\frac{\mathrm{rank}(G) - 12}{6} \right)^2 + \lambda_{\mathrm{gen}} \left(\frac{N_{\mathrm{gen}} - 3}{3} \right)^2 \\ + \lambda_Y m_Y^2 + \lambda_{\mathrm{Yuk}} \left(\frac{3 - \mathrm{rank}(Y_{\mathrm{Yukawa}})}{3} \right)^2 + \cdots . \end{aligned} \tag{39}$$

In a realistic construction, this term must be replaced by actual gauge group, chiral spectrum, hypercharge, Yukawa, and Higgs-sector data from branes, bundles, or singular geometries.

16.6 Cosmological and Observational Conditioning

The observational term factorises as

$$\mathcal{L}_{\text{obs}}(\mathcal{D} \mid \mathcal{B}) = \mathcal{L}_{\text{LE}}(D_{\text{LE}} \mid \mathcal{B})\, \mathcal{L}_{\Lambda}(D_{\Lambda} \mid \mathcal{B})\, \mathcal{L}_{\text{cos}}(D_{\text{cos}} \mid \mathcal{B})\, \mathcal{L}_{\text{anth}}(D_{\text{anth}} \mid \mathcal{B}). \tag{40}$$

The cosmological constant is therefore not claimed to be solved by definition. It is currently a conditioning factor. A stronger future result would show that low $\mathcal{F}_{\text{unified}}$ regions naturally concentrate near small positive Λ_{eff}.

16.7 Compact Master Statement

Combining the layers gives the compact master statement:

$$\boxed{\mathcal{B}_{\text{selected}} = \arg \min_{\mathcal{B} \in \mathfrak{B}_{\text{scan}}} \Big[\mathcal{F}_{\text{geom}} + \mathcal{F}_{\text{period}} + \mathcal{F}_{\text{KKLT}} + \mathcal{F}_{\text{back}} + \mathcal{F}_{\text{SM}} + \mathcal{F}_{\text{SW}} - \log W_{\text{cos}} - \log \mathcal{L}_{\text{obs}} \Big].} \tag{41}$$

In words:

> *A background is favoured when it is dynamically admissible, structurally coherent, period-controlled, KKLT-stabilised, weakly backreacted, compatible with SM-like low-energy data, not excluded by quantum-gravity consistency criteria, and compatible with cosmological and observational conditioning.*

This is the strongest form of the proposal developed here. It remains phenomenological, but it is no longer only verbal: every layer is associated with either a defined functional or an explicit benchmark proxy.

17 Code and Reproducibility

The executable benchmark scripts, JSON result files, and generated figures used in this paper are collected in the public repository under the `experiments/` directory. The repository entry point for reproducing the string-background selection benchmarks is:

`experiments/README.md in structural-selection-principle`.

The README lists the required Python packages, the execution order of the scripts, and the output files generated by each benchmark. The numerical pipeline is intentionally organised as a closed reproducibility loop:

model assumptions → generated ensemble → structural ranking → JSON results → plots.

18 What This Paper Addresses

This paper directly addresses the comparison problem:

Ingredient	**Role in the combined measure**
Swampland criteria	hard vetoes or soft consistency penalties
No-boundary weights	cosmological history weights or reference-measure factors
Anthropic reasoning	observational or environmental likelihoods
Structural selection	pre-anthropic prior ranking structural coherence
Low-energy data	phenomenological conditioning

It also gives a controlled place for the cosmological constant:

Λ enters as likelihood or environmental window,

not as a solved consequence of the current framework.

19 What Remains Open

The following points remain open and should not be hidden:

- A complete ten-dimensional or eleven-dimensional derivation of $\mathcal{F}_{\text{sel}}$ is not yet available.
- A realistic Standard-Model compactification has not yet been selected.
- The cosmological constant is conditioned on, not derived.
- Realistic quantitative predictions require global compactification ensembles beyond the toy, bridge, LCS, Picard–Fuchs, and backreaction/SM proxy scans introduced here.
- The relation between $\mathcal{F}_{\text{sel}}$ and exact quantum-gravity constraints is still effective and phenomenological.

These are not small gaps. They define the research programme.

20 Candidate Predictions

Even at the phenomenological level, the combined measure implies several testable expectations:

- structurally preferred vacua should cluster in regions with better EFT control and larger stability margins;
- energy-only ranking should fail to reproduce the full structural ranking in nontrivial ensembles;
- soft Swampland penalties and structural defects should be correlated but not identical;
- anthropically acceptable vacua should still split into structurally preferred and structurally fragile subsets;
- in KKLT-like benchmarks, tadpole closure and metastability should shift rankings even when vacuum energies are nearly degenerate.

Each claim can be tested in explicit scans.

21 Conclusion

This paper formulates a cautious phenomenological bridge between Structural Selection Theory and existing string-landscape selection ideas. The central claim is not that a final Theory of Everything has been completed. The claim is that landscape selection can be organised into distinct mathematical layers:

admissibility $\to$ structural prior $\to$ cosmological weight $\to$ observational conditioning.

The resulting master measure,

$$\mathrm{d}\mathbb{P}_{\text{pheno}}(\mathcal{B} \mid \mathcal{D}) \propto \chi_{\text{adm}}\, e^{-\mathcal{F}_{\text{sel}}-\mathcal{F}_{\text{SW}}}\, W_{\cos}\, \mathcal{L}_{\text{obs}}\, \mathrm{d}\mu_{\text{scan}},$$

does not solve every foundational problem. But it provides a precise arena in which Swampland constraints, no-boundary weights, anthropic conditioning, and structural selection can be compared without collapsing them into one another.

After the explicit benchmarks, this measure can be written in the consolidated form

$$\mathcal{B}_{\text{selected}} = \arg\min_{\mathcal{B}} \left[\mathcal{F}_{\text{geom}} + \mathcal{F}_{\text{period}} + \mathcal{F}_{\text{KKLT}} + \mathcal{F}_{\text{back}} + \mathcal{F}_{\text{SM}} + \mathcal{F}_{\text{SW}} - \log W_{\cos} - \log \mathcal{L}_{\text{obs}}\right].$$

This is the operational master formula of the present paper. It summarises the full chain from geometric data and Picard–Fuchs periods through KKLT stabilisation, backreaction proxies, SM-sector diagnostics, Swampland penalties, cosmological weighting, and observational conditioning.

This is the appropriate status of the proposal at present: not a final ToE, but a defined phenomenological selection measure over string backgrounds and a concrete path toward quantitative tests.

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